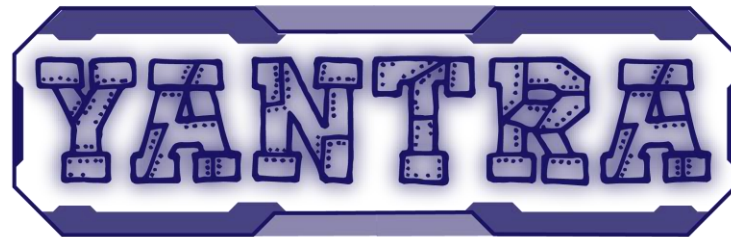


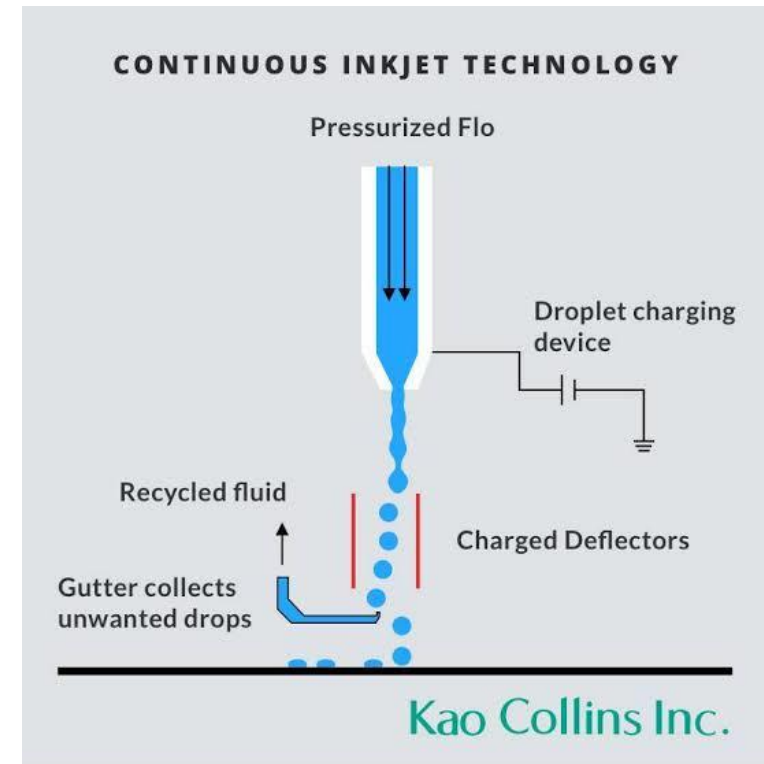
Concepts Modelers



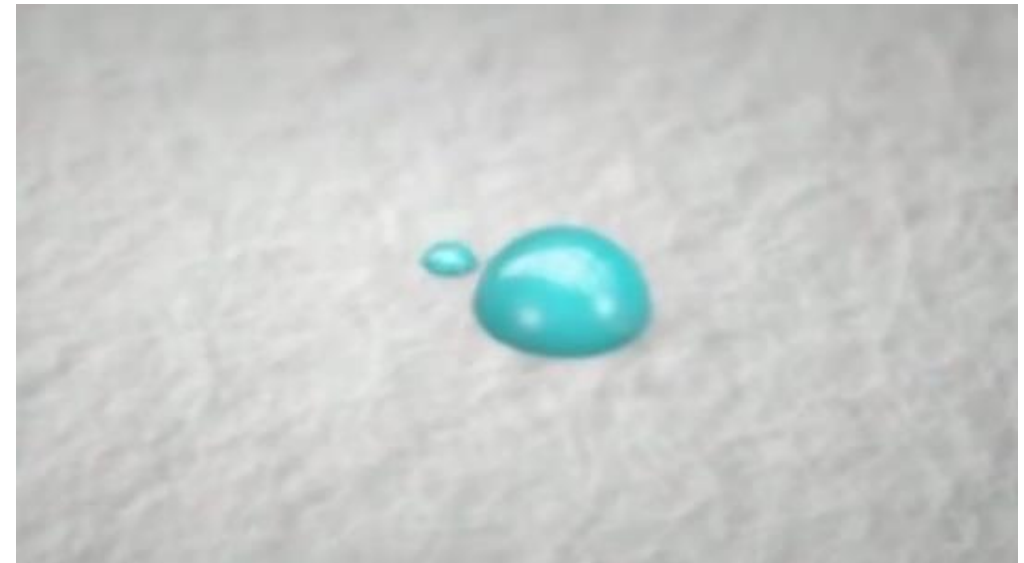
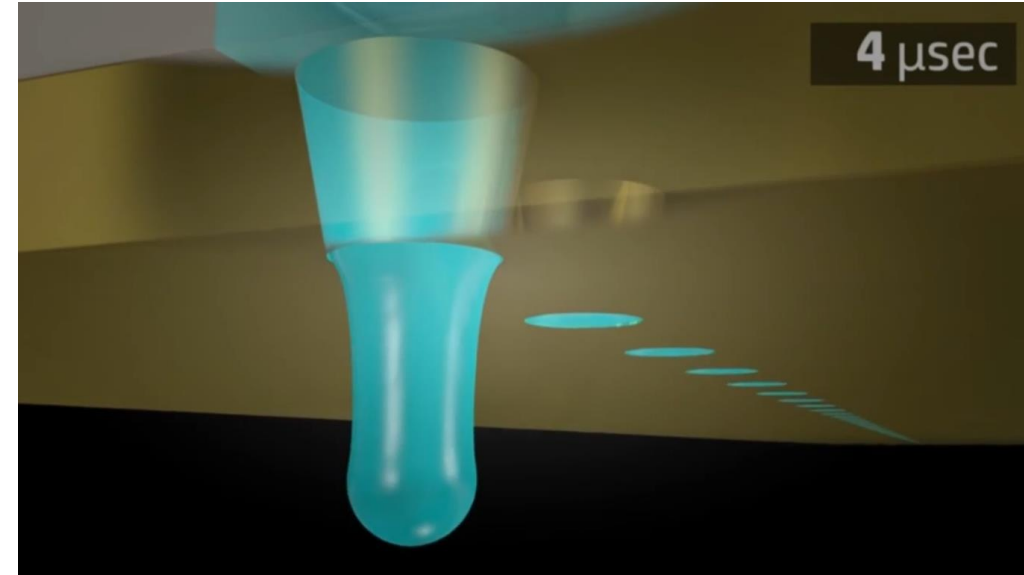
- Concept modelers, often called office modellers, are a class of rapid prototyping (RP) system
- designed specifically to make models **quickly** and **inexpensively**, without a great deal of effort.
- The systems are usually **small**, **inexpensive**, **quiet**, and require very little or no training to operate.
- For these reasons, the systems are targeted to reside in **design office environments**, where they can ideally be operated much like a standard printer, only the prints from these systems are in **three dimensions**.

Thermal Ink Jet Printer

- Ink jet printing comes from the printer and plotter industry where the technique involves **shooting tiny droplets of ink** on paper to produce graphic images.
- RP ink jet techniques utilize **ink jet technology** to shoot **droplets** of liquid-to-solid compound and form a layer of an RP model.
- The additive fabrication technique of inkjet printing is based on the 2D printer technique of using a jet to deposit tiny drops of ink onto paper.



- In Thermal Ink Jet Printer, the **ink is replaced** with **thermoplastic** and **wax** materials, which are held in a melted state.
- When printed, **liquid drops** of these materials instantly **cool and solidify** to form a layer of the part.
- For this reason, the process is often referred to as **thermal phase change** inkjet printing. Inkjet printing offers the advantages of excellent accuracy and surface finishes.

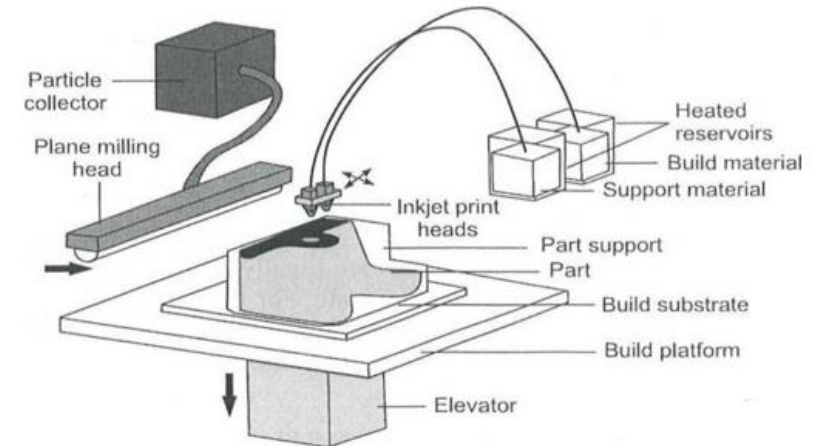
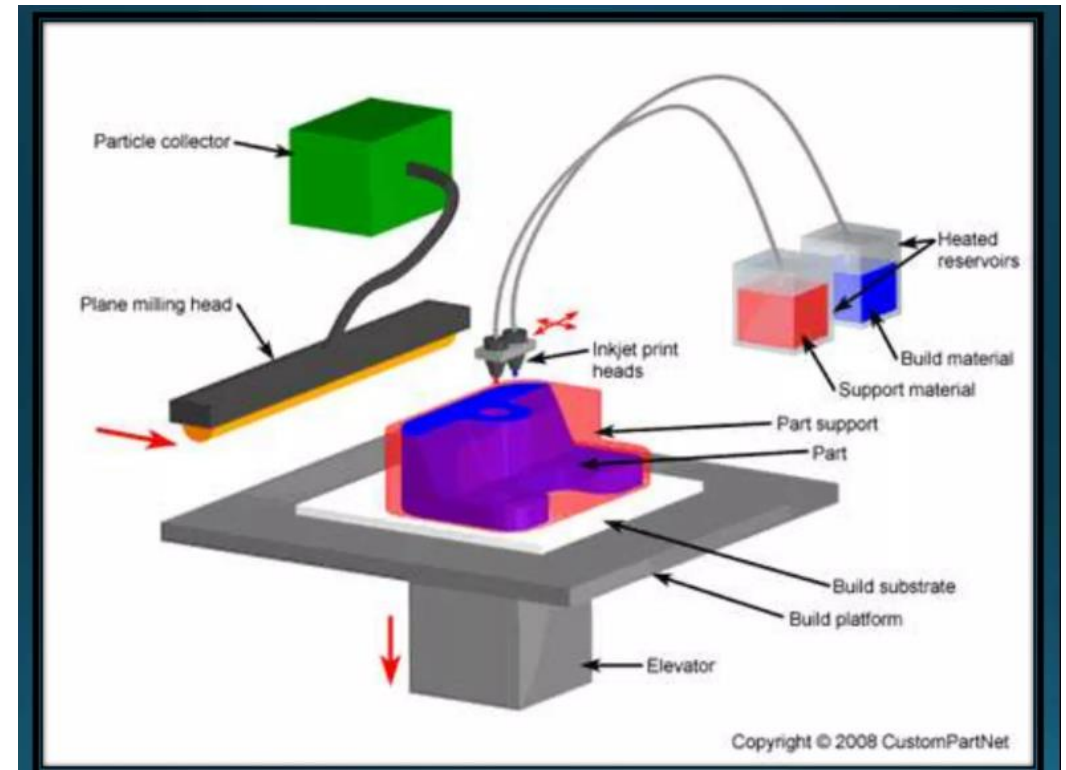


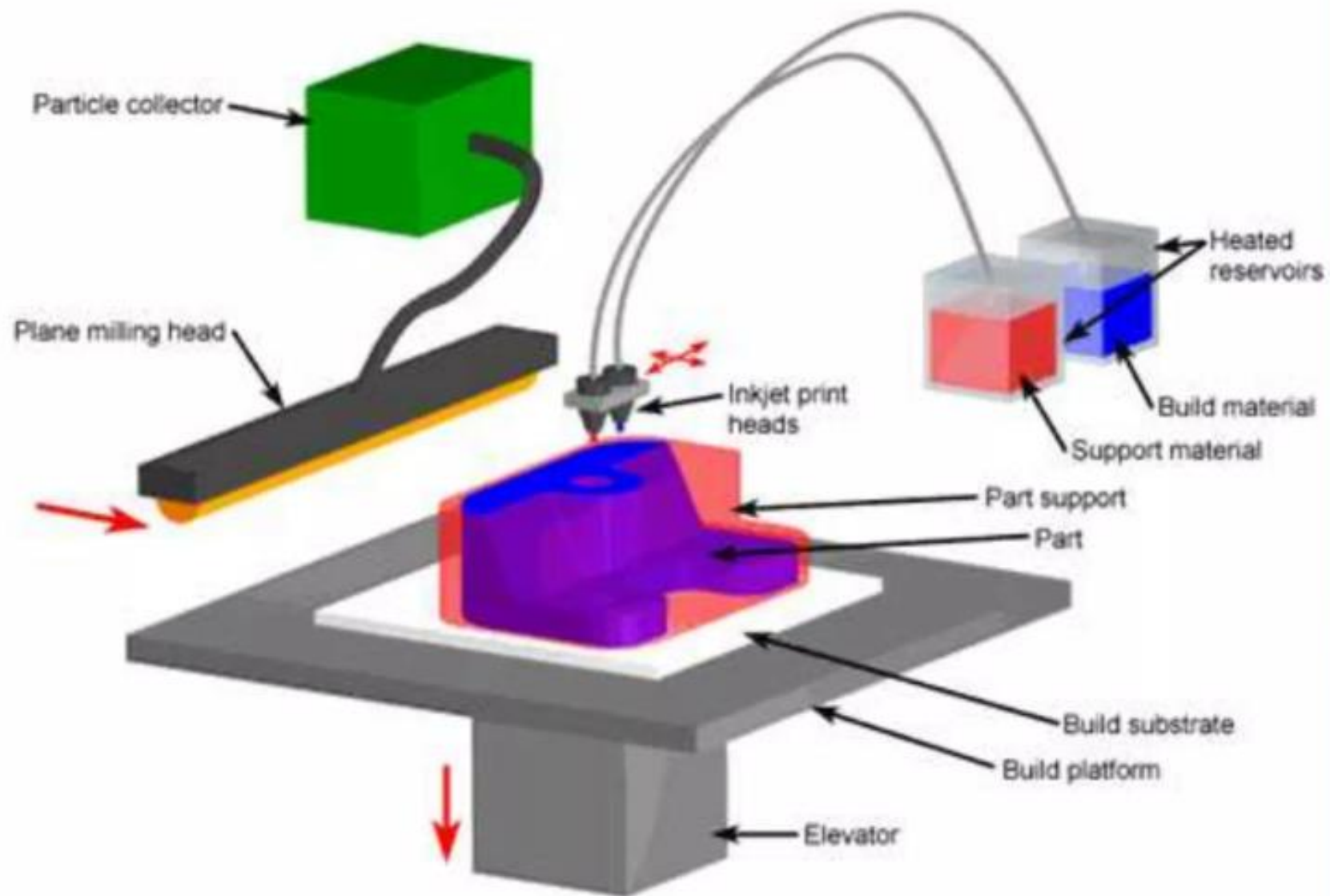
- However, the **limitations** include slow build speeds, few material options, and fragile parts.
- Thus, the most **common application** of inkjet printing is prototypes used for form and fit testing.
- Inkjet printers from Solidscape Inc., such as the Sander's Model Maker, use a single jet for the build material and another jet for support material.
- Multi-Jet Modeling (MJM) techno ThermoJet Modeller machines that utilize **several hundred** nozzles to enable faster build times.

The inkjet printing process, begins with the **build material** (thermoplastic) and **support material** (wax) being held in a melted state inside two heated reservoirs.

These materials are each fed to an inkjet print head which moves in the **X-Y plane** and shoots tiny droplets to the required locations to form one layer of the part.

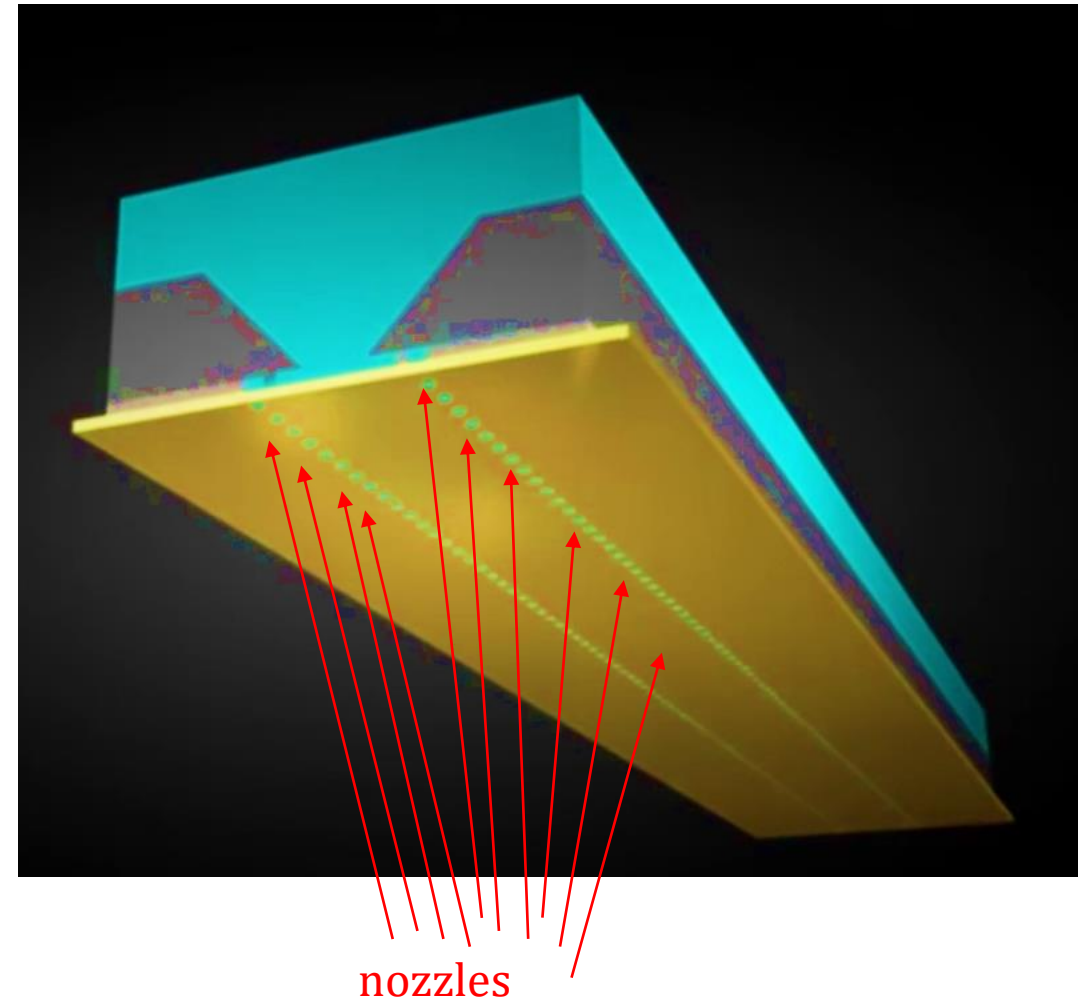
Both the build material and support material **instantly cool and solidify**.

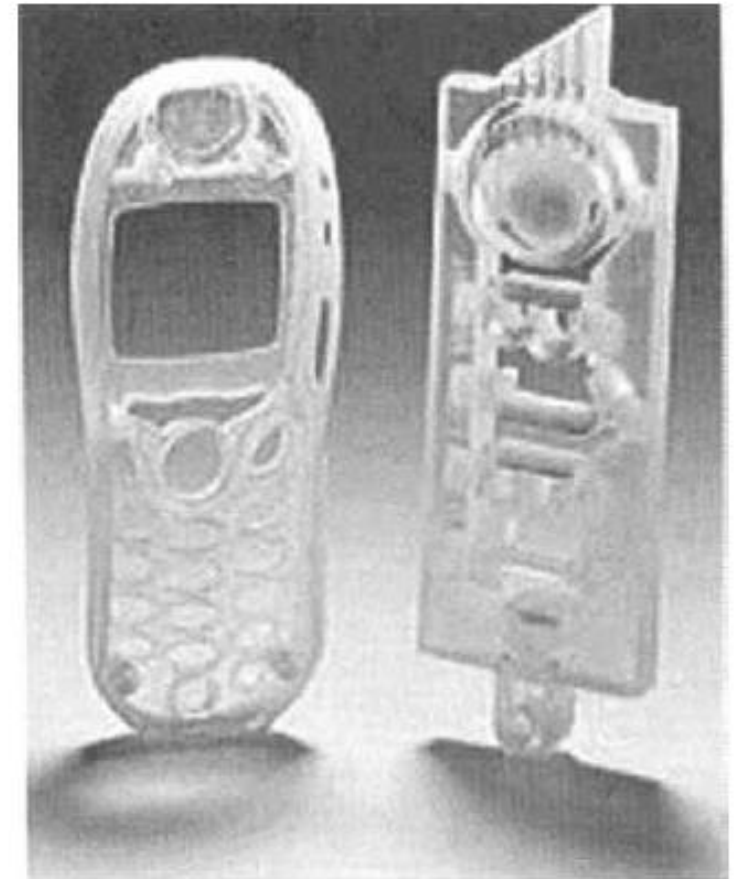
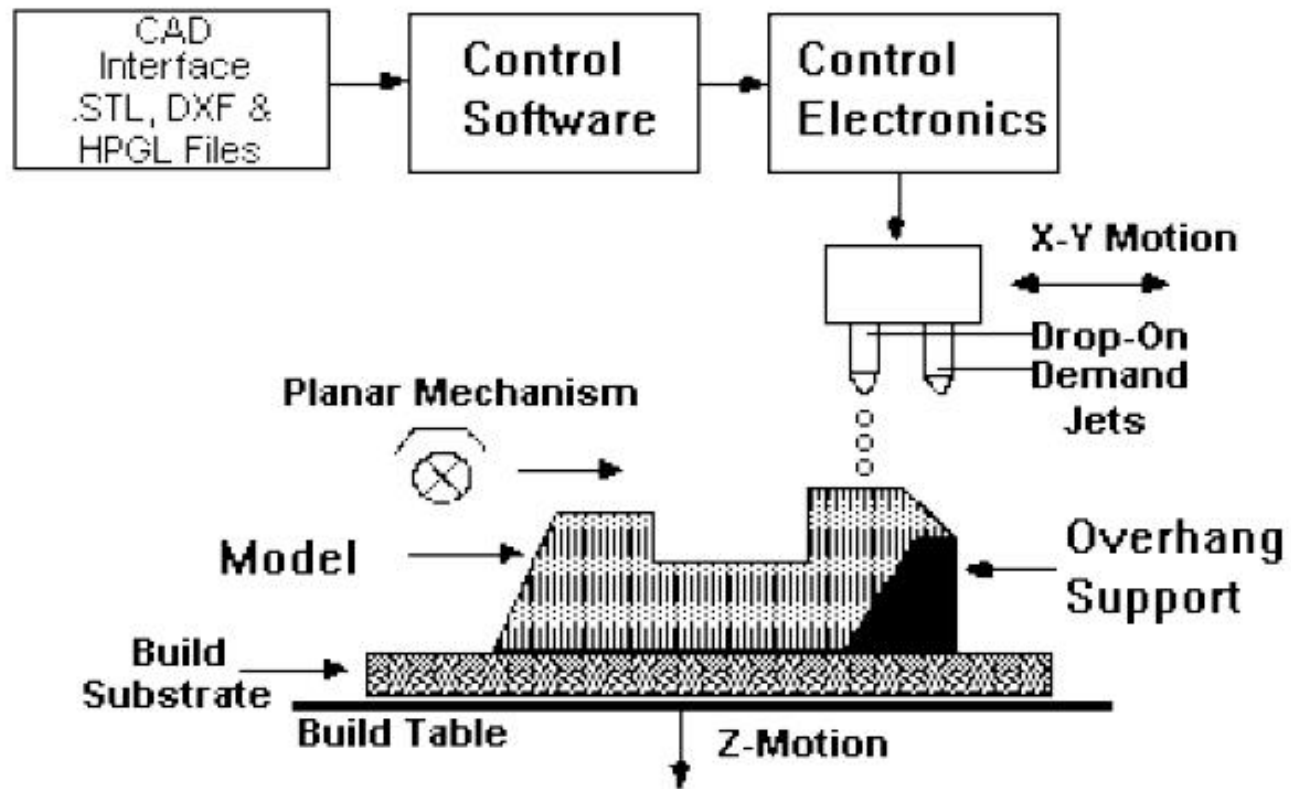




The Object Quadra

- The Quadra process is based on state-of-the-art **ink-jet printing technology**.
- The printer, which uses **1536** nozzles, jets a **proprietary photopolymer** developed in-house by Object.
- Because it requires **no post-cure or postprocessing**, Quadra touts the fastest start-to-finish process of any (RP) machine currently on the market.

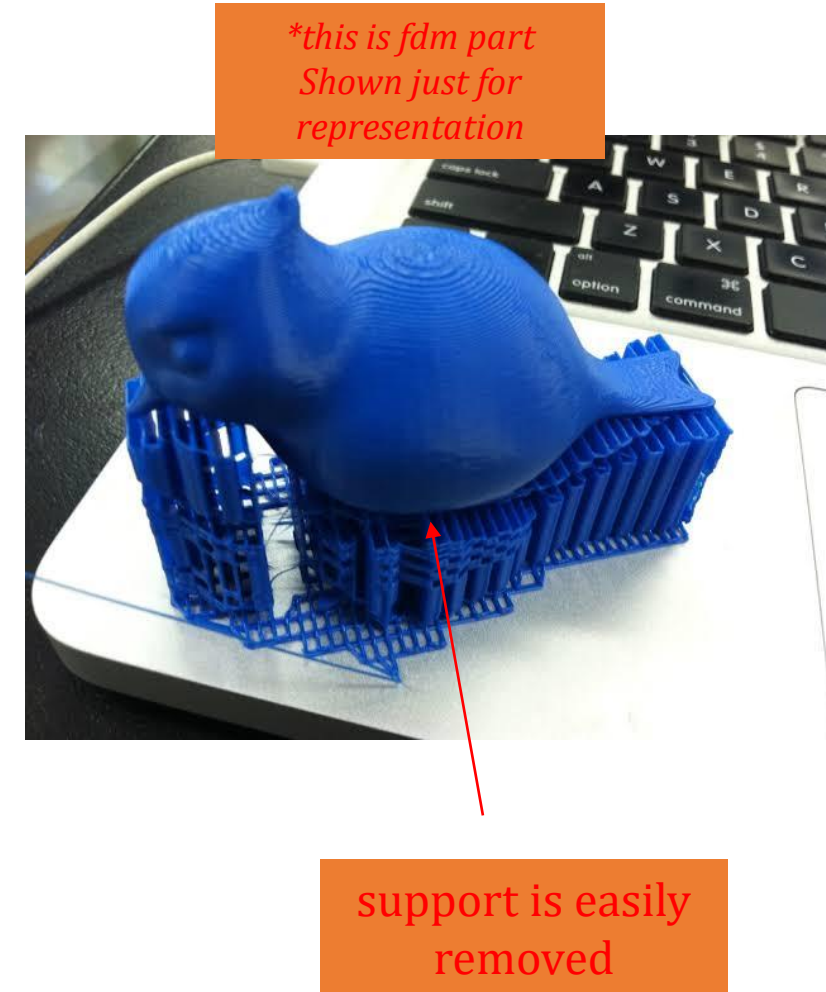




- Material is delivered by a **sealed cartridge** that can be easily installed and replaced.
- Jetting of different resins, once they become available, & does not require costly investments in materials or hardware upgrades.
- A new cartridge can be **dropped** into place without any complicated procedures or specially trained staff.
- Quadra can deposit **second material** that is jetted to **support** models containing complicated geometry, such as overhangs and undercuts.



- The support material is can be **easily removed** by hand after building the model.
- The support material **separates** easily from the model body without leaving any **contact points** or blemishes to the model. No special staff or training are required.
- Furthermore, models built on the system do not require sanding or smoothing where the supports are attached.



- The Quadra prints in a resolution of **600 dpi**, with a layer thickness of 20 microns, and builds parts up to a maximum size of 11" x 12" x 8".
- The introduction of Quadra marks the start of a revolution in three-dimensional imaging.
- An **intuitive user interface** aids users in setting up the build, scaling, and positioning single and multiple models.

